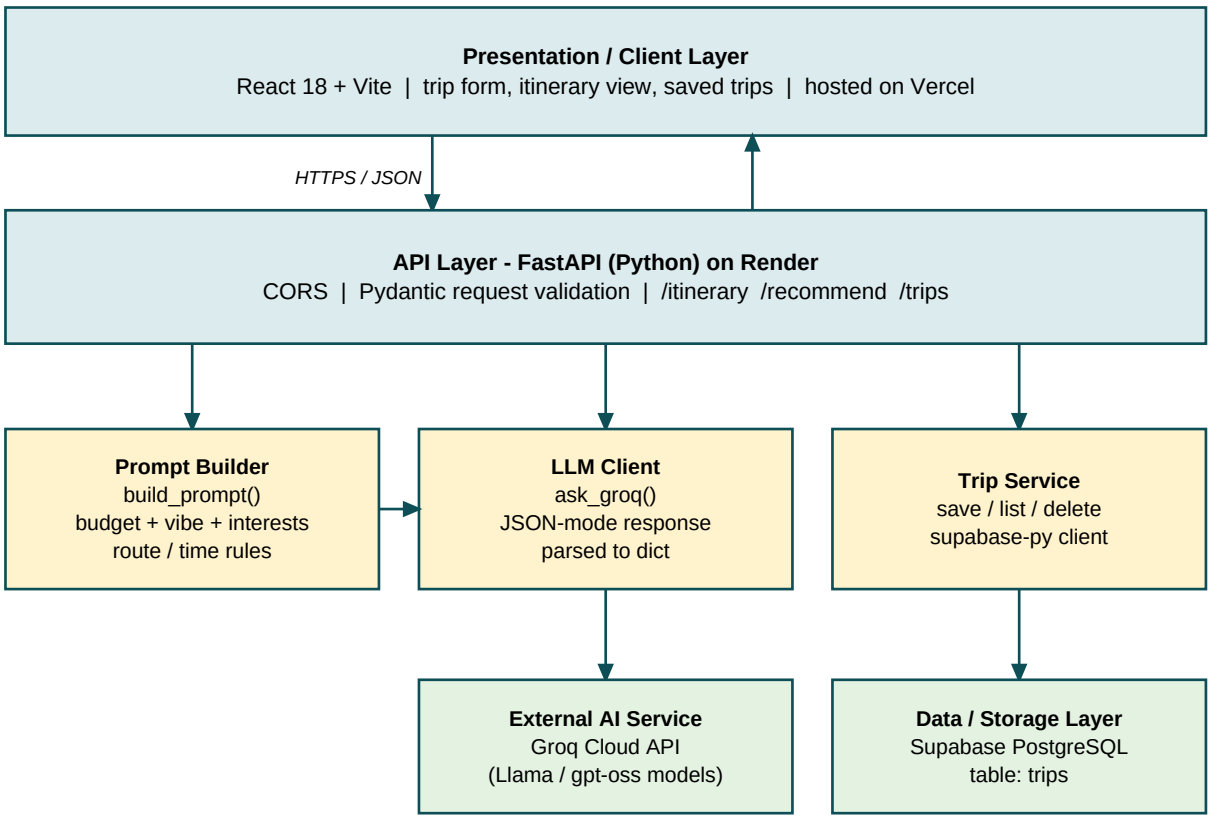
Solution Architecture

Date	20 October 2026
Team ID	Individual project (no team)
Project Name	Personalized Travel Itinerary Planner
Maximum Marks	5 Marks

Solution Architecture Diagram



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Trip form, itinerary view, saved trips list, copy-as-text; calls the backend API.	React 18, Vite, CSS, hosted on Vercel
API Layer	Receives requests, validates input, applies CORS, routes to the right service, returns JSON.	FastAPI, Pydantic, Uvicorn on Render
Prompt Builder	Turns preferences into a structured prompt with route, time, cost and vibe rules.	Python (build_prompt)
LLM Client	Sends the prompt to Groq and parses the JSON response.	Groq API, JSON mode (ask_groq)
Trip Service	Saves, lists and deletes trips in the database.	supabase-py
Database	Stores trip rows with preferences and the full itinerary as JSONB.	Supabase PostgreSQL (table: trips)